Generalized Linear Models.

	Linear.	Poisson	Binary Logistic R	Multiclass Logistic. R
Distribution of Y, given $X_1 \cdots X_p$.	Gaussian.	Poisson.	Bernoulli	Categorical.
mean of the distribution = $E[Y \mid X_1 \cdots X_p] =$ Let $\eta = \beta_0 + \beta_1 X_1 + \cdots + X_p$	$= \eta$. (model)	$= \lambda(X_1 \cdots X_p)$. $= e^{\eta}$ (model)	$= Pr(Y=1 \mid X_1 \cdots X_p)$. $= \frac{e^{\eta}}{1+e^{\eta}}$. (model). $= \frac{1}{1+e^{-\eta}}$ (logistic function)	=
link function $\ell(\underbrace{E[Y \mid X_1 \cdots X_p}_{\text{mean}}) = \eta$. link mean to η (eta)	$g(X) = X$. $g(\eta) = \eta$.	$g(X) = \log(X)$. $g(e^{\eta}) = \log e^{\eta} = \eta$	$g(X) = \log \frac{X}{1-X}$. $g\left(\frac{1}{1+e^{-\eta}}\right) = \log \frac{\frac{e^{\eta}}{1+e^{\eta}}}{1-\frac{e^{\eta}}{1+e^{\eta}}}$ $= \log \frac{e^{\eta}}{1+e^{\eta}-e^{\eta}} = \log e^{\eta} = \eta$.	